# Cloud Security Operations & Incident Response

## Centralised Logging · Flow Logs · Automated Alerting · Threat Detection · Dashboards · IR Playbook

*Prepared by Sabitha S Karanavar · Azure subscription "Azure subscription 1" · October 2026 · Working draft*

## 1. Scope and AWS → Azure mapping

The rubric names AWS services (CloudTrail, VPC Flow Logs, GuardDuty). HVMC runs on Microsoft Azure, so each requirement is met with the native Azure equivalent, deployed into the existing resource group `HVMC-CloudSecurity-RG` (East US) and the HVMC-Prod-VNet three-tier network.

| Rubric / AWS service | HVMC Azure implementation | What it records or does | Coverage |
|---|---|---|---|
| **AWS CloudTrail (management events)** | Azure Activity Log -> diagnostic setting HVMC-ActivityLog-to-LAW | Every control-plane write/delete/action in the subscription: who, what, when, from which IP, result | 8 categories |
| **AWS CloudTrail (data events)** | Resource logs: StorageRead/Write/Delete on hvmcehrstorage | Every read, write and delete of an ePHI blob, with caller identity and IP (HIPAA 164.312(b)) | ePHI store |
| **CloudTrail (session audit)** | BastionAuditLogs on HVMC-Bastion | Who opened an admin session to which host, and when | Admin access |
| **VPC Flow Logs** | VNet flow log HVMC-Prod-VNet-FlowLog + Traffic Analytics | Allowed/denied 5-tuple flows for every NIC in the VNet, 30-day retention, analysed every 10 min | 4 subnets |
| **Security-group logs** | NetworkSecurityGroupEvent / RuleCounter on the 3 NSGs | Which NSG rule matched how often | 3 NSGs |
| **CloudWatch Logs** | Log Analytics workspace HVMC-SecOps-LAW | Single queryable store (KQL), 30-day retention, 0.5 GB/day cap | Central |
| **CloudWatch Alarms / EventBridge + SNS** | 9 activity-log alert rules -> action group HVMC-SecOps-AG (e-mail) | Near-real-time notification of security-relevant changes | 9 rules |
| **AWS Budgets** | Consumption budget HVMC-Lab-Budget ($10/month, 50/80/100 %) | Cost anomaly warning | Cost guard |

## 2. Step 1 — Centralised logging, flow logs and automated alerting

The whole step is a single, re-runnable script (`scripts/01_setup_logging_alerting.sh`) so the configuration is documented as code and can be rebuilt or removed (`99_teardown.sh`). It was run on 15 October 2026 against the live subscription. Figure 1 shows the run.

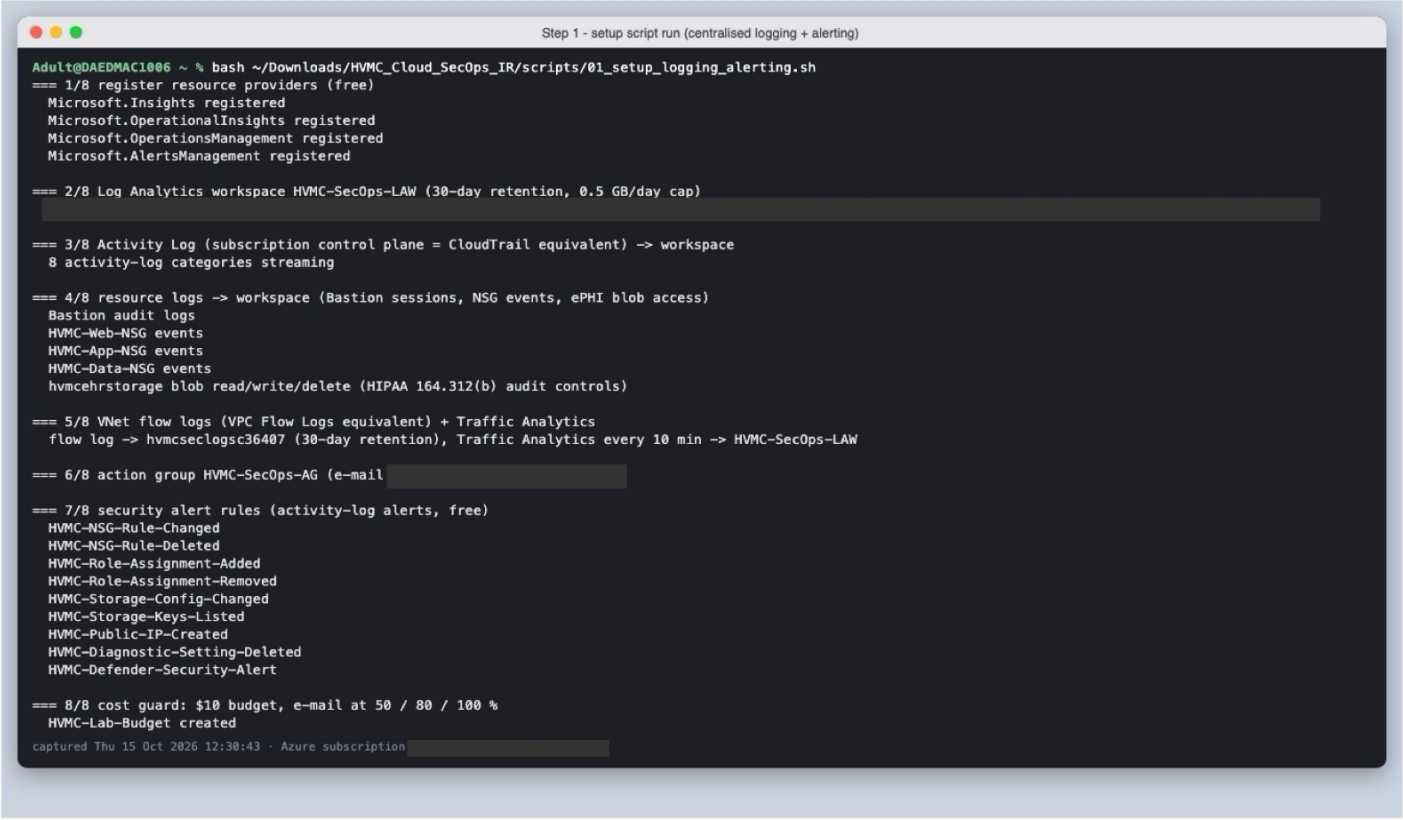



*Figure 1. Setup script run: providers registered, workspace created, logs connected, flow logs, action group, 9 alert rules and budget.*

## 2.1 Central log store

All security telemetry lands in one Log Analytics workspace, so a single KQL query can join who changed a firewall rule (Activity Log) with what traffic followed (flow logs). Retention is 30 days (inside the free retention period) and a 0.5 GB/day ingestion cap protects the free-trial credit.

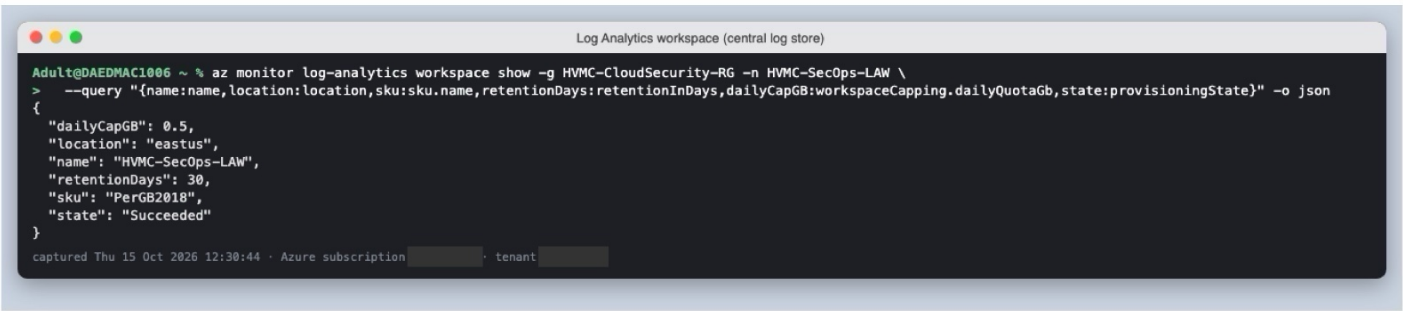



*Figure 2. Log Analytics workspace HVMC-SecOps-LAW: PerGB2018, 30-day retention, 0.5 GB/day cap.*

## 2.2 CloudTrail equivalent — Activity Log and resource audit logs

The subscription Activity Log is exported with all eight categories (Administrative, Security, ServiceHealth, Alert, Recommendation, Policy, Autoscale, ResourceHealth). Resource logs add the data-plane detail that the Activity Log does not hold: Bastion session audit, NSG rule events, and every read/write/delete on the ePHI storage account — the HIPAA §164.312(b) audit-control requirement that was flagged as a gap in the HIPAA analysis. Together with the alert rules below, this step treats risk **R01** in the HVMC risk register — “Security incident goes undetected for months (no central logging or alerting)”, the top-rated risk — whose finding was 0 workspaces, 0 activity-log exports, 0 alert rules and 0 flow logs.

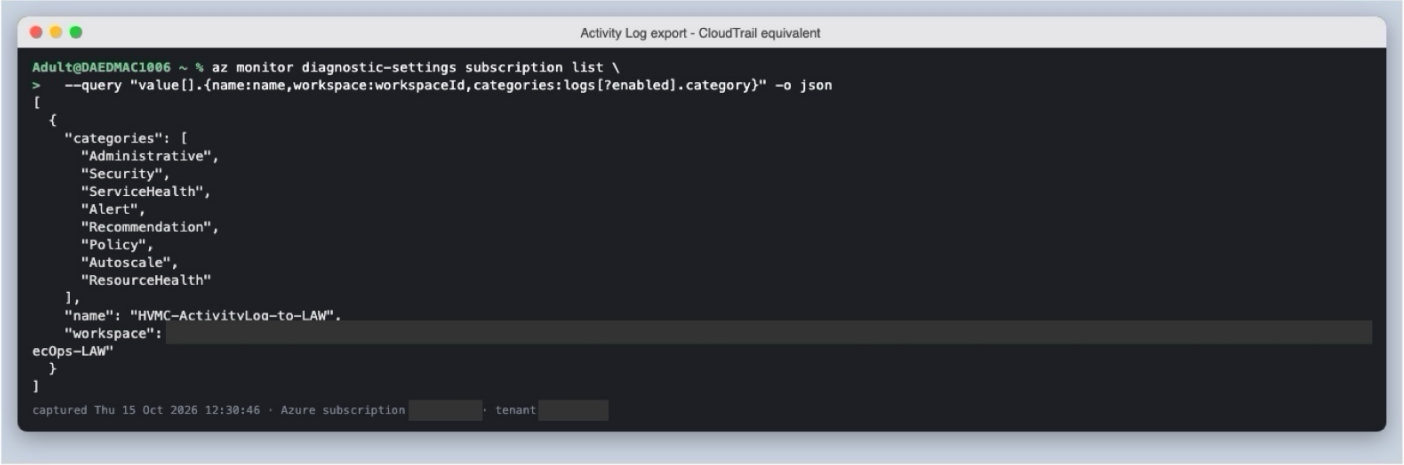



*Figure 3. Subscription Activity Log exported to the workspace with all eight categories.*

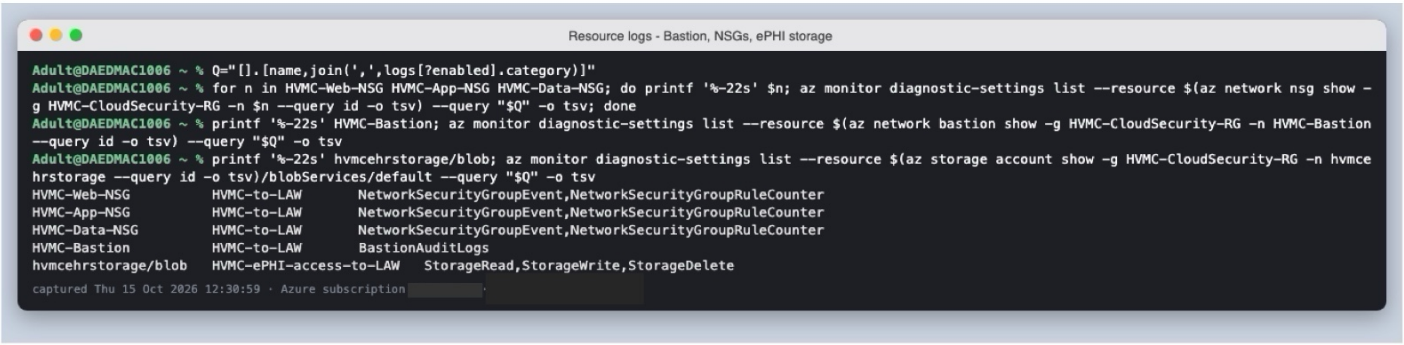



*Figure 4. Resource logs: NSG events on all three tiers, Bastion audit logs, and ePHI blob access on hvmcehrstorage.*

## 2.3 VPC Flow Logs equivalent — VNet flow logs

A VNet-level flow log (the successor to NSG flow logs, which Azure stopped accepting for new deployments in 2025) records every flow through HVMC-Prod-VNet into a dedicated log storage account kept separate from the ePHI account, with 30-day retention. Traffic Analytics processes the flows every 10 minutes into the workspace. *Note:* no virtual machines run in the VNet, so flow volume is low (Bastion and load-balancer traffic only); by 14:35 local time on 15 October Traffic Analytics had already written 99 flow records (NTANetAnalytics) and 139 IP-detail records to the workspace (see the Workbook figure in section 4).

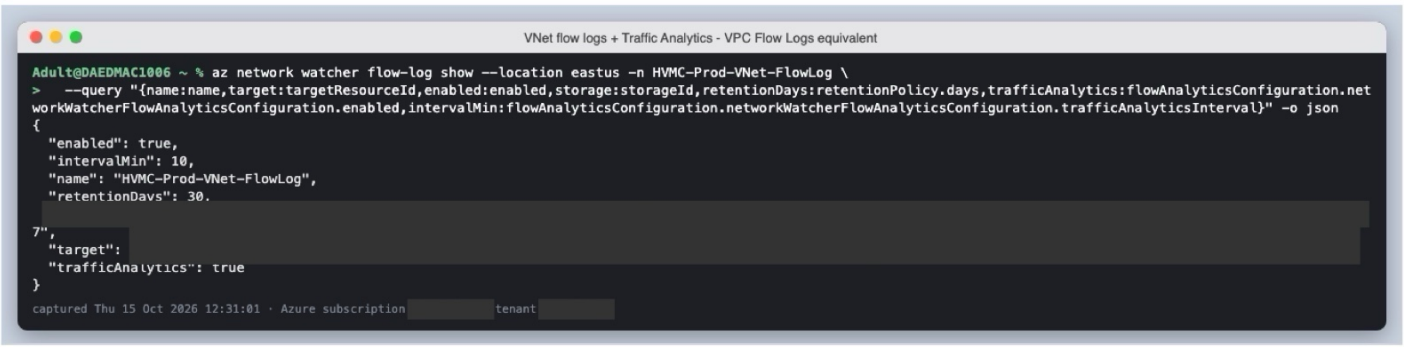



*Figure 5. VNet flow log on HVMC-Prod-VNet with 30-day retention and Traffic Analytics every 10 minutes.*

## 2.4 Automated alerting

Nine activity-log alert rules watch for the changes that matter most for HVMC's risk profile. Each rule is mapped to a risk in the register, an incident response playbook and a MITRE ATT&CK technique, and notifies the Security Lead by e-mail through the action group HVMC-SecOps-AG (common alert schema).

| Alert rule | Trigger (operation) | Risk | Playbook | Threat / ATT&CK |
|---|---|---|---|---|
| **HVMC-NSG-Rule-Changed** | `securityRules/write` | R03 | PB-01 Internet exposure | Initial access (T1190 enabler) |
| **HVMC-NSG-Rule-Deleted** | `securityRules/delete` | R03 | PB-01 | Defence evasion |
| **HVMC-Role-Assignment-Added** | `roleAssignments/write` | R04 | PB-02 Privilege escalation | T1098 Account manipulation |
| **HVMC-Role-Assignment-Removed** | `roleAssignments/delete` | R04 | PB-02 | Impact / cover tracks |
| **HVMC-Storage-Config-Changed** | `storageAccounts/write` | R02 | PB-03 ePHI exposure | T1530 Data from cloud storage |
| **HVMC-Storage-Keys-Listed** | `storageAccounts/listKeys/action` | R02, R05 | PB-03 | T1552 Unsecured credentials |
| **HVMC-Public-IP-Created** | `publicIPAddresses/write` | R03, R10 | PB-01 | New attack surface |
| **HVMC-Diagnostic-Setting-Deleted** | `diagnosticSettings/delete` | R01 | PB-04 Logging tampering | T1562.008 Disable cloud logs |
| **HVMC-Defender-Security-Alert** | `category = Security` | R11, all | per alert type | Any Defender detection |

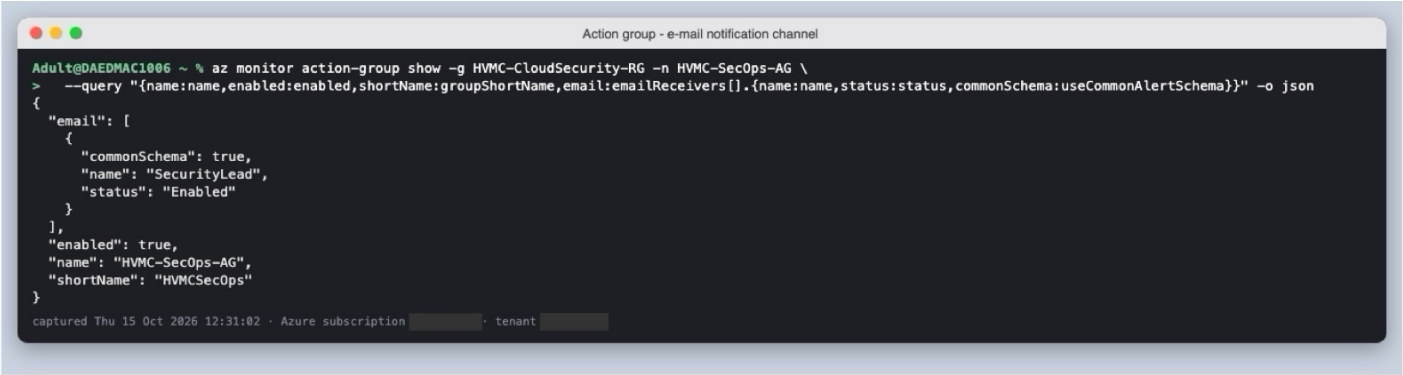



*Figure 6. Action group HVMC-SecOps-AG: e-mail receiver for the Security Lead.*

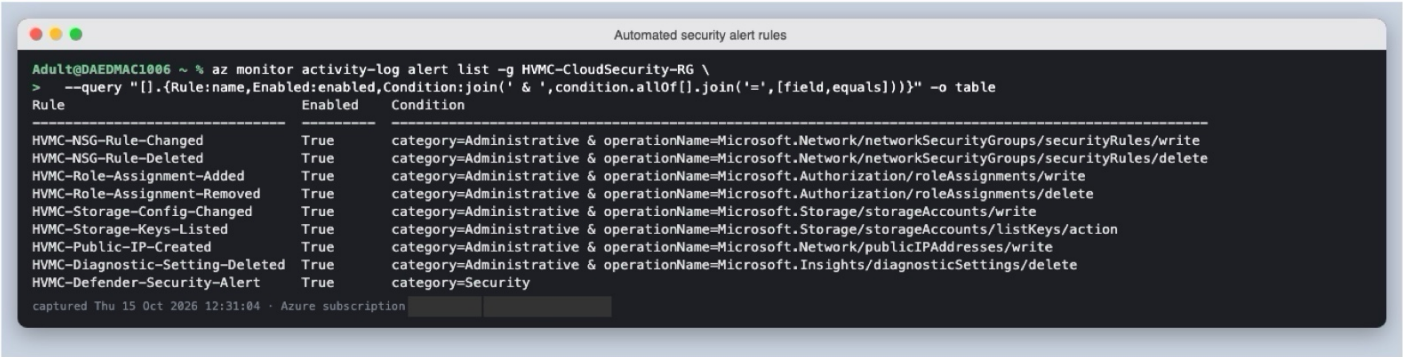



*Figure 7. The nine automated security alert rules, all enabled.*

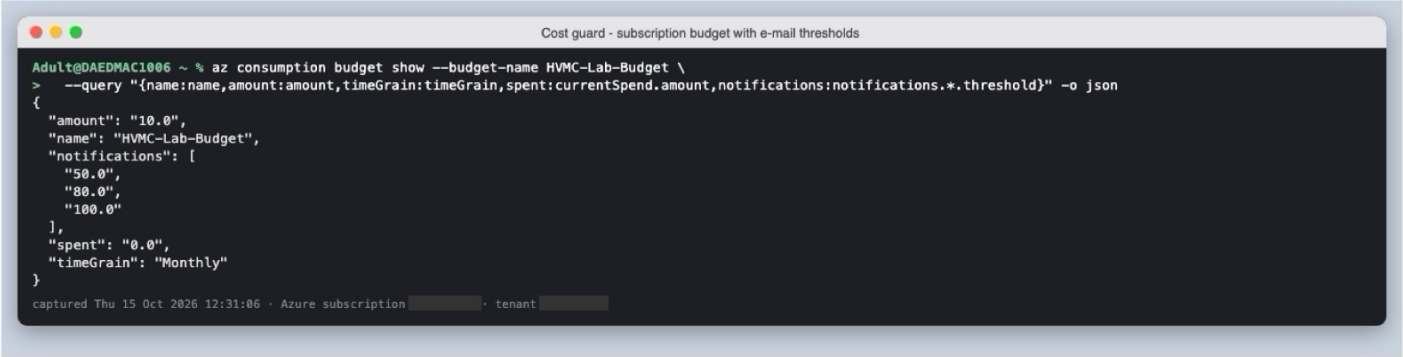



*Figure 8. Cost guard: $10 monthly budget with e-mail warnings at 50, 80 and 100 percent.*

## 3. Step 2 — Threat detection (GuardDuty equivalent)

Logging and rule-based alerts (Step 1) only catch what was anticipated. Step 2 adds two layers that look for *suspicious behaviour*, the role GuardDuty plays in AWS, deployed with `scripts/02_setup_threat_detection.sh`:

| Layer | Service | What it detects for HVMC | Cost |
|---|---|---|---|
| **Cloud-native threat detection** | Microsoft Defender for Storage (Defender for Storage V2) | Unusual access to the ePHI account (access from Tor / unusual location, anomalous data extraction, suspicious SAS use) and **on-upload malware scanning** of every new blob, with the result written as a blob index tag. Risks R02, R05, **R11**. | 30-day free trial; scanning capped at 1 GB/month/account ($0.15/GB) |
| **Control-plane threat detection** | Microsoft Defender for Resource Manager | Suspicious management operations: known attacker toolkits (MicroBurst, PowerZure), access from malicious IPs, unusual permission or resource changes. Risks R04, R08. | 30-day free trial |
| **SIEM / correlation** | Microsoft Sentinel on HVMC-SecOps-LAW | Correlates the Step 1 logs and the Defender alerts; analytics rules turn detections into tracked **incidents** with owner, status and entities (account, IP, resource). | 31-day free trial (10 GB/day) |
| **Notification** | Defender security contact | E-mails the Security Lead for every Defender alert of Medium severity or higher. | Free |

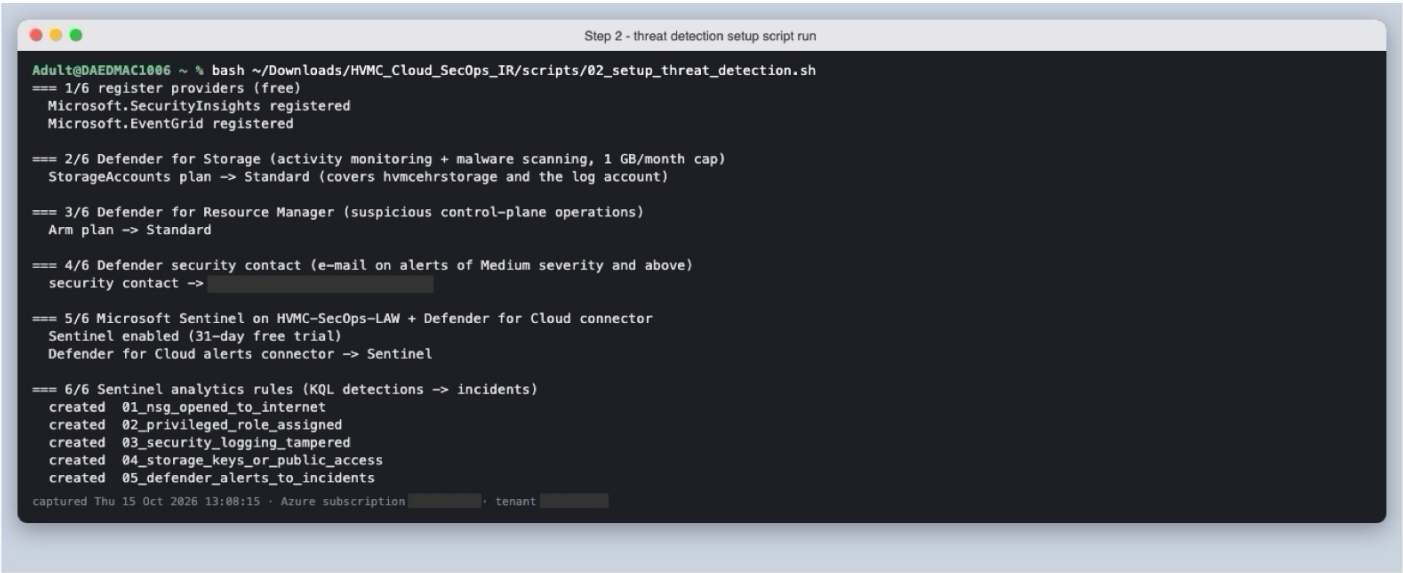



*Figure 9. Step 2 setup script run: Defender plans, security contact, Sentinel, connector and five analytics rules created.*

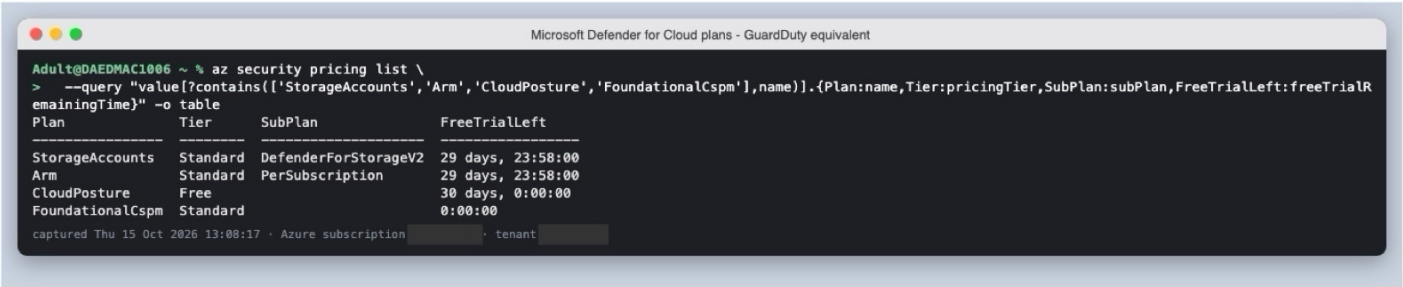



*Figure 10. Defender for Cloud: Storage and Resource Manager plans on Standard with the 30-day free trial running.*

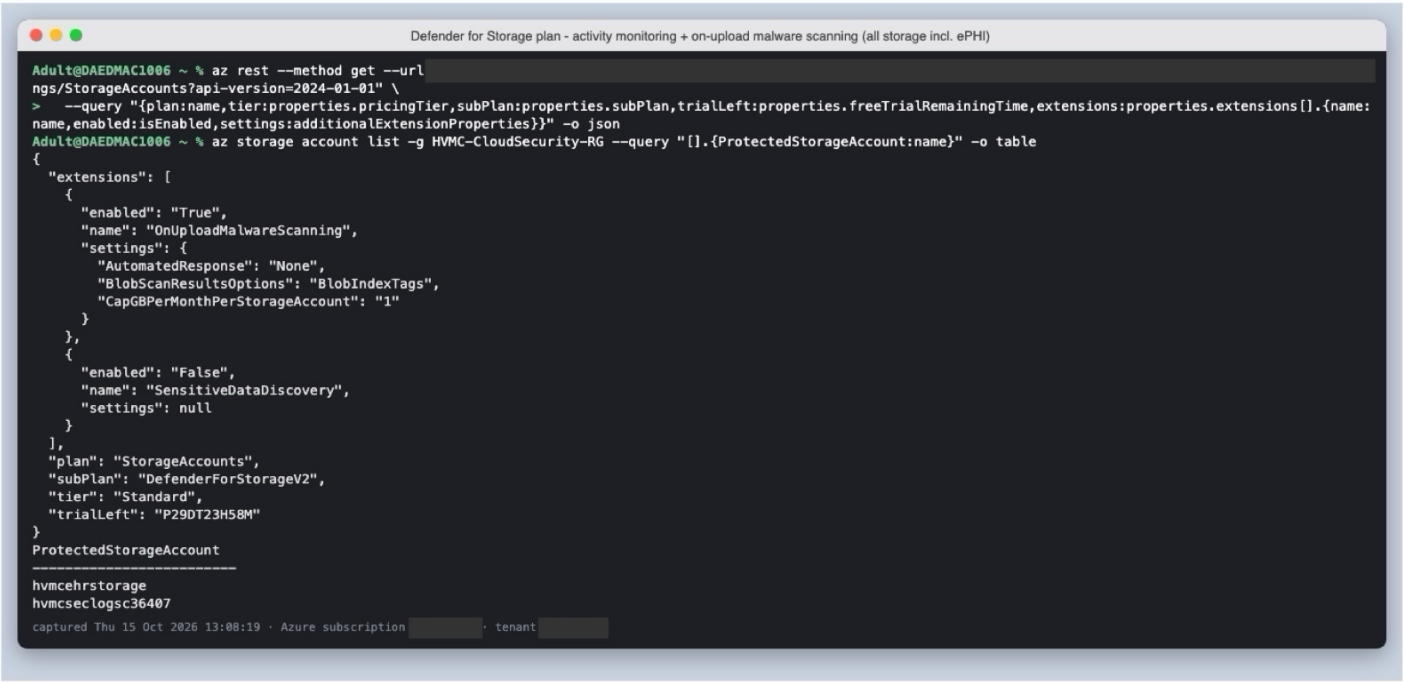



*Figure 11. Defender for Storage plan settings: malware scanning on upload (1 GB/month cap) protecting both storage accounts, including the ePHI store.*

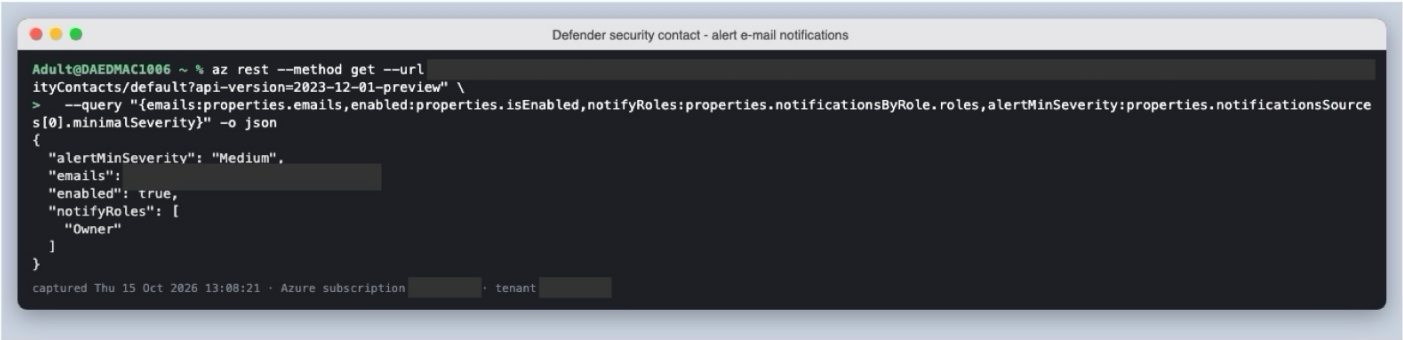



*Figure 12. Defender security contact: e-mail to the Security Lead and subscription Owners for Medium+ alerts.*

## 3.1 Sentinel analytics rules

Four custom scheduled KQL rules run every five minutes over a 15-minute window, and a Microsoft-security rule turns every Defender alert into an incident. Sentinel also enables its built-in **Fusion** rule, which correlates low-fidelity signals into multistage-attack incidents. The custom rules go further than the Step 1 alerts: they read the request body, so a firewall change only becomes a High incident when it really *allows inbound traffic from the Internet*, and a role assignment only when the role is Owner, Contributor or User Access Administrator. Each query was tested against the live workspace before deployment, and each rule is stored as JSON in `scripts/sentinel_rules/`.

| Rule | Severity | MITRE ATT&CK | Risk | Playbook |
|---|---|---|---|---|
| NSG inbound rule opened to the Internet | High | Initial Access, Defense Evasion (T1190, T1562) | R03, R10 | PB-01 |
| Privileged Azure role assigned | High | Privilege Escalation, Persistence (T1098) | R04, R07 | PB-02 |
| Security logging disabled or deleted (incl. Defender plan downgraded) | High | Defense Evasion (T1562) | R01 | PB-04 |
| ePHI storage keys retrieved or storage exposure changed | Medium | Credential Access, Collection (T1552, T1530) | R02, R05 | PB-03 |
| Incidents from Defender for Cloud alerts | per alert | per alert | R11 and others | by alert type |

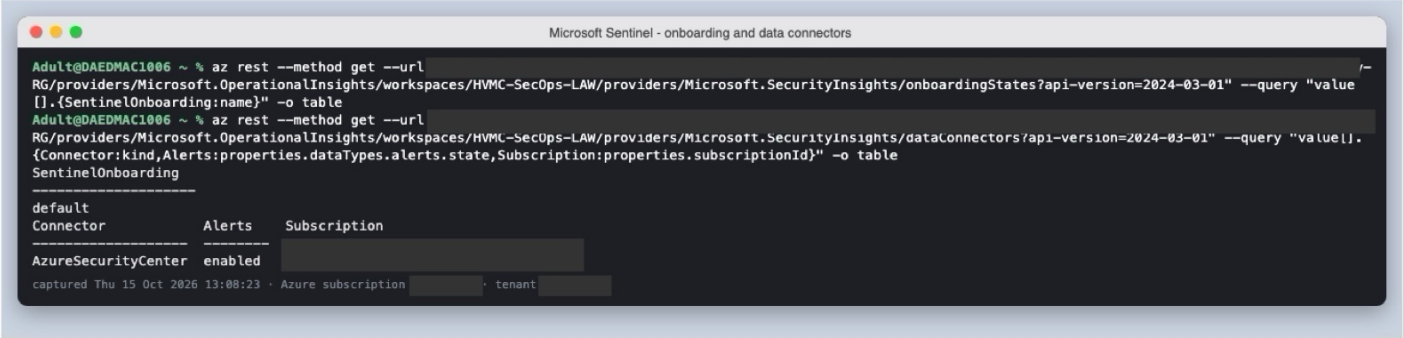



*Figure 13. Microsoft Sentinel onboarded on HVMC-SecOps-LAW with the Defender for Cloud alerts connector enabled.*

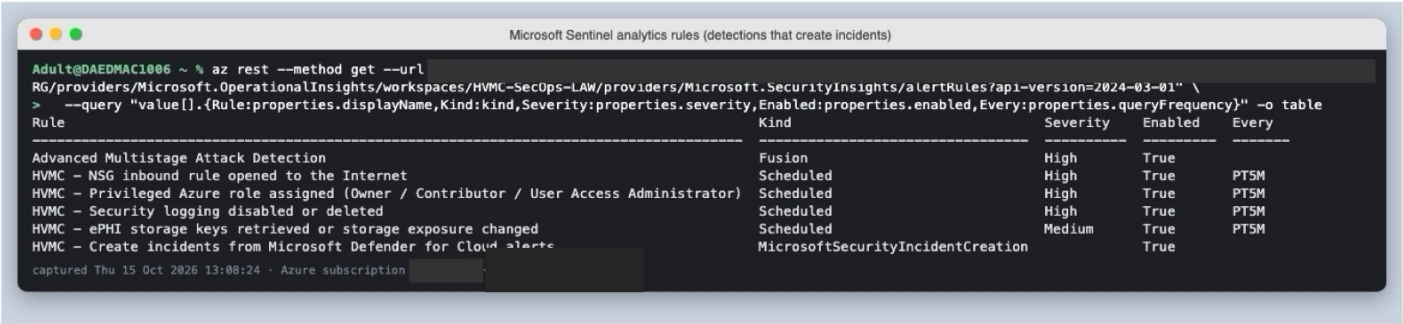



*Figure 14. Sentinel analytics rules: four HVMC scheduled detections, the Defender incident rule and the built-in Fusion rule.*

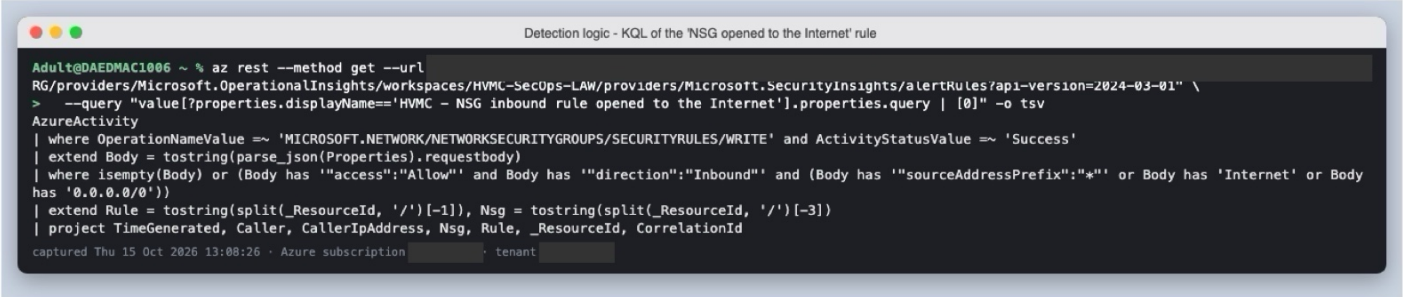



*Figure 15. Detection logic of the 'NSG inbound rule opened to the Internet' rule (KQL).*

*Not included: Microsoft Entra ID sign-in and audit logs. Exporting them to a workspace requires an Entra ID P1/P2 licence, which the lab tenant does not have; they remain viewable in the Entra portal for 7 days. This is recorded as a known gap (risk R12, AiTM phishing).*

## 4. Step 3 — Security dashboards and key metrics

Two dashboards present the same telemetry for different audiences. Both are free: they only read data that is already collected.

| Dashboard | Audience | Built with | How it is refreshed |
|---|---|---|---|
| **HVMC Security Operations Dashboard** (HTML) | Security Lead, IT Manager, auditors — one page, no Azure login needed | `dashboard/collect_metrics.py` (read-only Azure CLI + KQL) → `build_dashboard.py` ; triage notes in `triage.json` | Re-run the two scripts (same pattern as the KRI monitor from the risk work); light and dark mode, hover tooltips |
| **HVMC Security Operations** Azure Workbook | SOC analyst working inside Azure / Sentinel | `scripts/03_create_workbook.sh` , 10 KQL tiles with a time-range picker ( `dashboard/workbook_queries.py` ) | Live: each tile queries HVMC-SecOps-LAW when opened |

### 4.1 Key metrics

| Metric | Definition | Source | Target / threshold | Value 15 Oct 2026, after the exercise |
|---|---|---|---|---|
| **Log source coverage** | Security log sources connected ÷ sources required | Diagnostic settings, flow log, Defender plans, Sentinel | 100 %; any “Missing” = Red | 10 / 11 (Entra ID logs: licence gap) |
| **Detection rules enabled** | Enabled alert + analytics rules | Azure Monitor, Sentinel | All enabled; a disabled rule = Red (PB-04) | 15 / 15 |
| **Alerts fired / untriaged** | Alerts in 7 days; alerts without a disposition | Azure Monitor alerts + triage log | Untriaged = 0 within 1 business day | 40 fired · 0 untriaged |
| **Open incidents · MTTA · MTTR** | Open Sentinel incidents; mean minutes created → first action → closed | SecurityIncident | High: acknowledge ≤ 15 min, contain ≤ 1 h | 0 open · 8 closed (4 duplicates after grouping fix) |
| **Security-relevant changes** | Firewall, role, storage, key, public-IP, logging and Defender changes | AzureActivity | Each must match an approved change (IP-1) | 10 (all from approved changes) |
| **Defender secure score** | Defender for Cloud secure score | Defender for Cloud | ≥ 80 % by Phase 2 of the risk program | 50 % (6 / 12) |
| **Ingestion vs cap** | MB ingested per day vs 0.5 GB cap | Usage | < 80 % of cap | < 1 MB/day |
| **Credit used** | Trial credit consumed, by resource type | Cost Management | SecOps tooling = $0 | $53.35 total · SecOps $0.00 |

### 4.2 First real detections and triage

The alerting worked before any test was run. Eleven alerts fired during setup and were triaged within the hour; both groups were authorised changes, so they are recorded as benign with evidence (the dashboard below also shows the 29 alerts from the Step 4 exercise):

- **16:32 UTC — HVMC-Storage-Config-Changed (×3):** the Step 1 script created the flow-log storage account. Caller verified (Sabitha S Karanavar via Azure CLI); the ePHI account was not touched.
- **17:12 UTC — HVMC-Role-Assignment-Added (×8):** enabling Defender for Storage and Sentinel caused Azure to grant roles to Microsoft service principals — *Defender for Storage Scanner Operator* and *EventGrid Contributor* at subscription scope, and *User Access Administrator* scoped only to the Sentinel workspace. These are expected platform identities; recording them stops them being mistaken for privilege escalation later (lesson learned: pre-register planned changes so their alerts can be auto-closed).

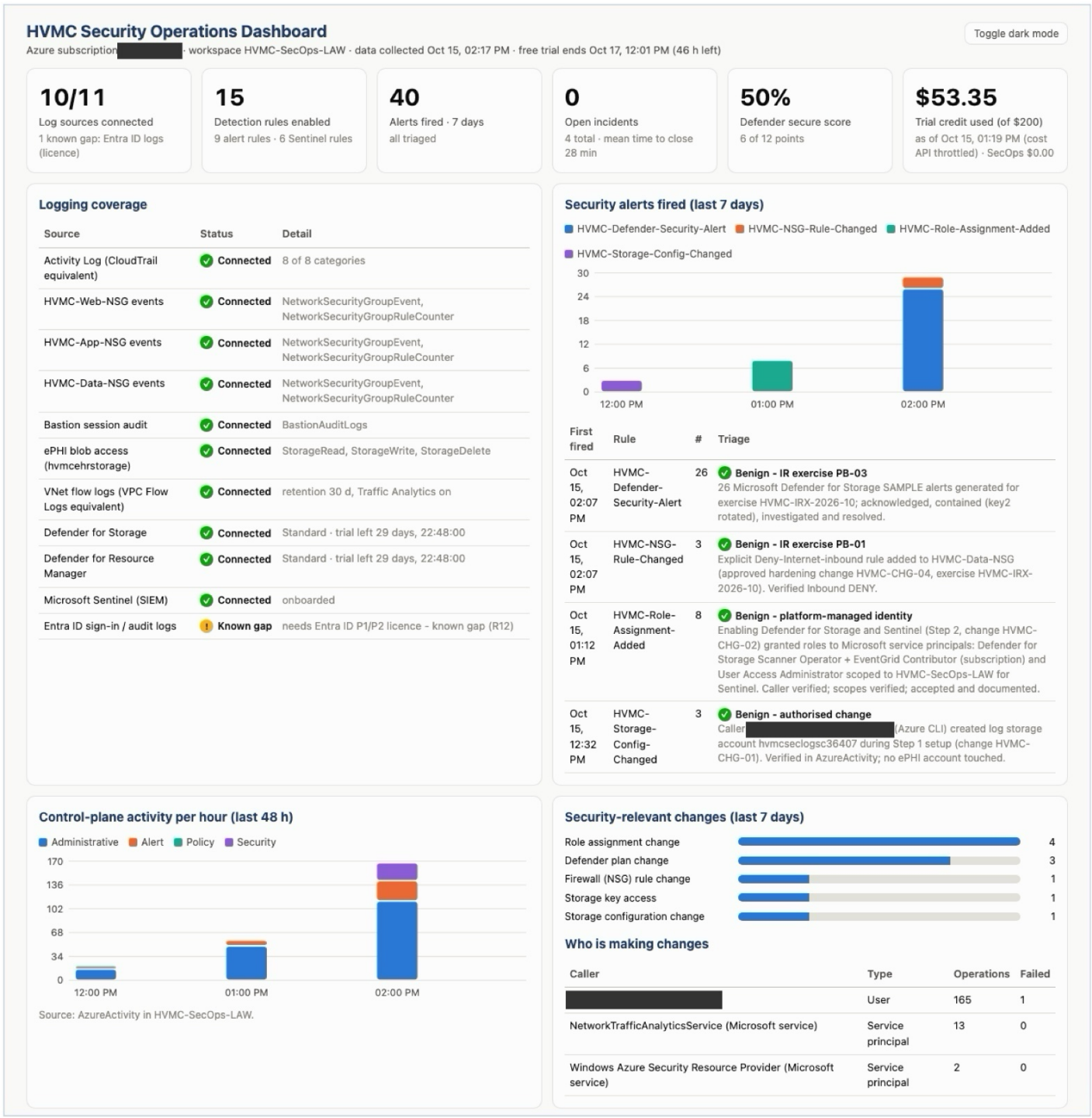



*Figure 16. HVMC Security Operations Dashboard (top): KPI tiles, logging coverage, alerts fired with triage dispositions, activity and changes.*

### Incidents (Microsoft Sentinel)

| # | Title | Severity | Status | Created | Time to close |
|---|---|---|---|---|---|
| 1 | HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator) | High | Closed | Oct 15, 01:17 PM | 57 min |
| 4 | HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction | Medium | Closed | Oct 15, 02:15 PM | 0 min |
| 3 | HVMC - NSG inbound rule opened to the Internet | High | Closed | Oct 15, 02:12 PM | 2 min |
| 2 | HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator) | High | Closed | Oct 15, 01:22 PM | 52 min |

### Detection rules

| Rule | Type | Severity | Status |
|---|---|---|---|
| HVMC-NSG-Rule-Changed | Activity-log alert | — | Enabled |
| HVMC-NSG-Rule-Deleted | Activity-log alert | — | Enabled |
| HVMC-Role-Assignment-Added | Activity-log alert | — | Enabled |
| HVMC-Role-Assignment-Removed | Activity-log alert | — | Enabled |
| HVMC-Storage-Config-Changed | Activity-log alert | — | Enabled |
| HVMC-Storage-Keys-Listed | Activity-log alert | — | Enabled |
| HVMC-Public-IP-Created | Activity-log alert | — | Enabled |
| HVMC-Diagnostic-Setting-Deleted | Activity-log alert | — | Enabled |
| HVMC-Defender-Security-Alert | Activity-log alert | — | Enabled |
| Advanced Multistage Attack Detection | Sentinel Fusion | High | Enabled |
| HVMC - NSG inbound rule opened to the Internet | Sentinel Scheduled | High | Enabled |
| HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator) | Sentinel Scheduled | High | Enabled |
| HVMC - Security logging disabled or deleted | Sentinel Scheduled | High | Enabled |
| HVMC - ePHI storage keys retrieved or storage exposure changed | Sentinel Scheduled | Medium | Enabled |
| HVMC - Create incidents from Microsoft Defender for Cloud alerts | Sentinel MicrosoftSecurityIncidentCreation | per alert | Enabled |

### Security posture (Defender for Cloud)

**50%** secure score · 6 / 12

5 healthy · 10 unhealthy · 13 not applicable. Defender re-evaluates about every 24 h, so items fixed in Step 2 may still be listed.

| Open recommendation | Status |
|---|---|
| Email notification to subscription owner for high severity alerts should be enabled | Unhealthy |
| Microsoft Defender for Resource Manager should be enabled | Unhealthy |
| Email notification for high severity alerts should be enabled | Unhealthy |
| Subscriptions should have a contact email address for security issues | Unhealthy |
| Microsoft Defender CSPM should be enabled | Unhealthy |
| Storage account should use a private link connection | Unhealthy |
| Storage accounts should restrict network access using virtual network rules | Unhealthy |
| Storage accounts should prevent shared key access | Unhealthy |
| Microsoft Defender for Storage plan should be enabled with Malware Scanning and Sensitive Data Threat Detection | Unhealthy |
| There should be more than one owner assigned to subscriptions | Unhealthy |

### Cost guard (trial credit used)

**$53.35** of $200 free-trial credit · spending limit on, no card charge

| | |
|---|---|
| bastionhosts | $39.58 |
| natgateways | $10.41 |
| publicipaddresses | $3.36 |
| loadbalancers | $0.00 |
| storageaccounts | $0.00 |

All credit use comes from the pre-existing Bastion, NAT gateway and public IPs. Security-operations tooling (workspace, alerts, Defender and Sentinel trials) has cost $0.00 so far. Cost data can lag by up to 24 h.

Read-only data collected by collect_metrics.py · triage dispositions from triage.json · Horizon Valley Medical Center · Sabitha S Karanavar

*Figure 17. HVMC Security Operations Dashboard (bottom): incidents, detection rules, Defender posture and cost guard.*

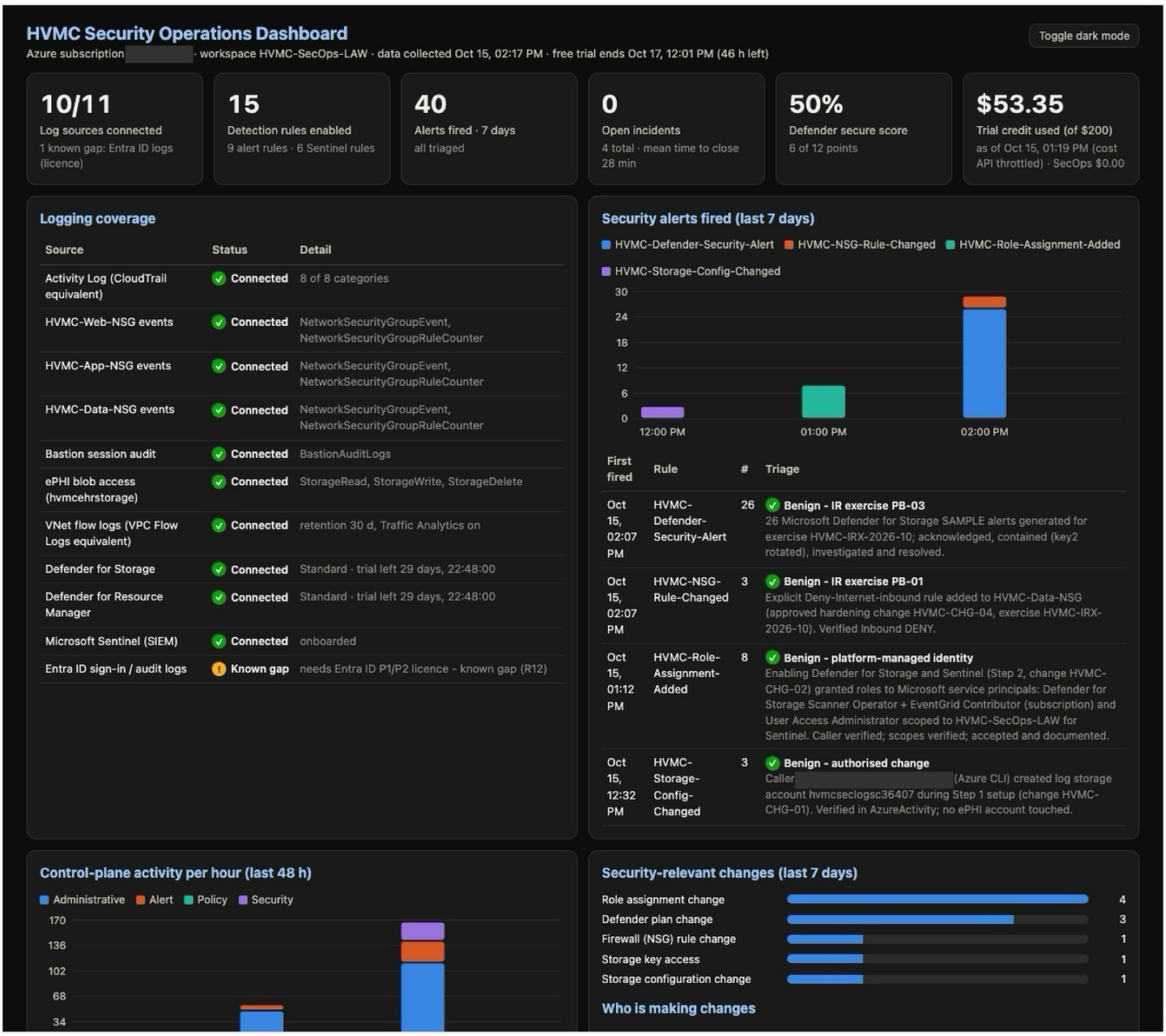



*Figure 18. The same dashboard in dark mode (each colour step re-chosen for the dark surface, not inverted).*

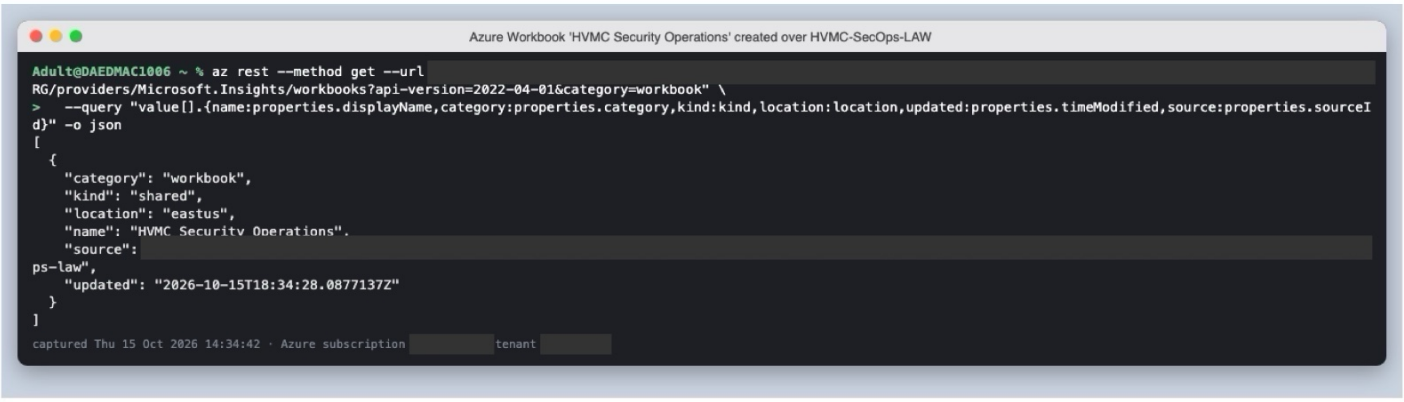



*Figure 19. Azure Workbook 'HVMC Security Operations' deployed by scripts/03_create_workbook.sh, linked to HVMC-SecOps-LAW.*

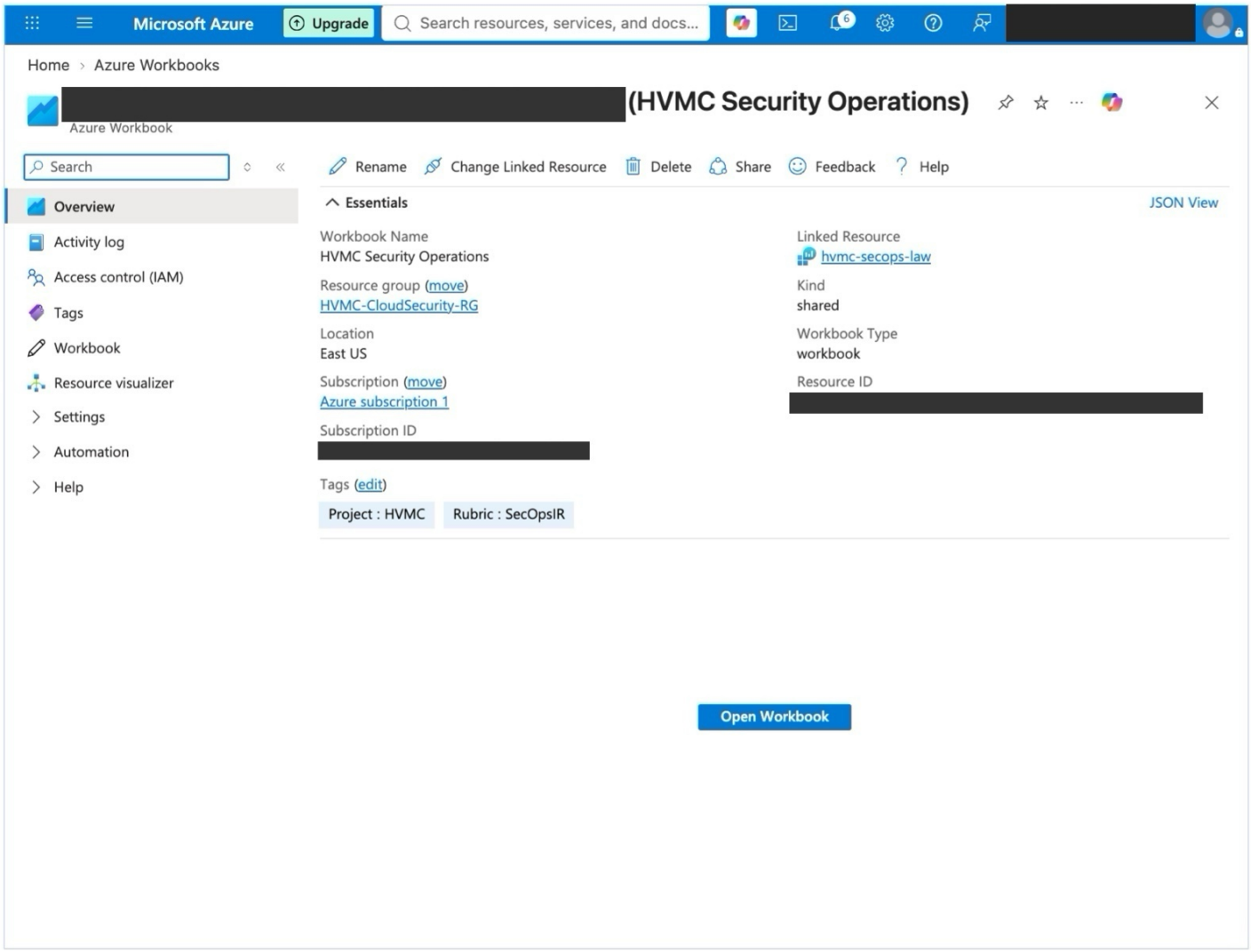



*Figure 20. The workbook resource in the Azure portal: linked to hvmc-secops-law, HVMC-CloudSecurity-RG, tagged Project HVMC / Rubric SecOpsIR.*

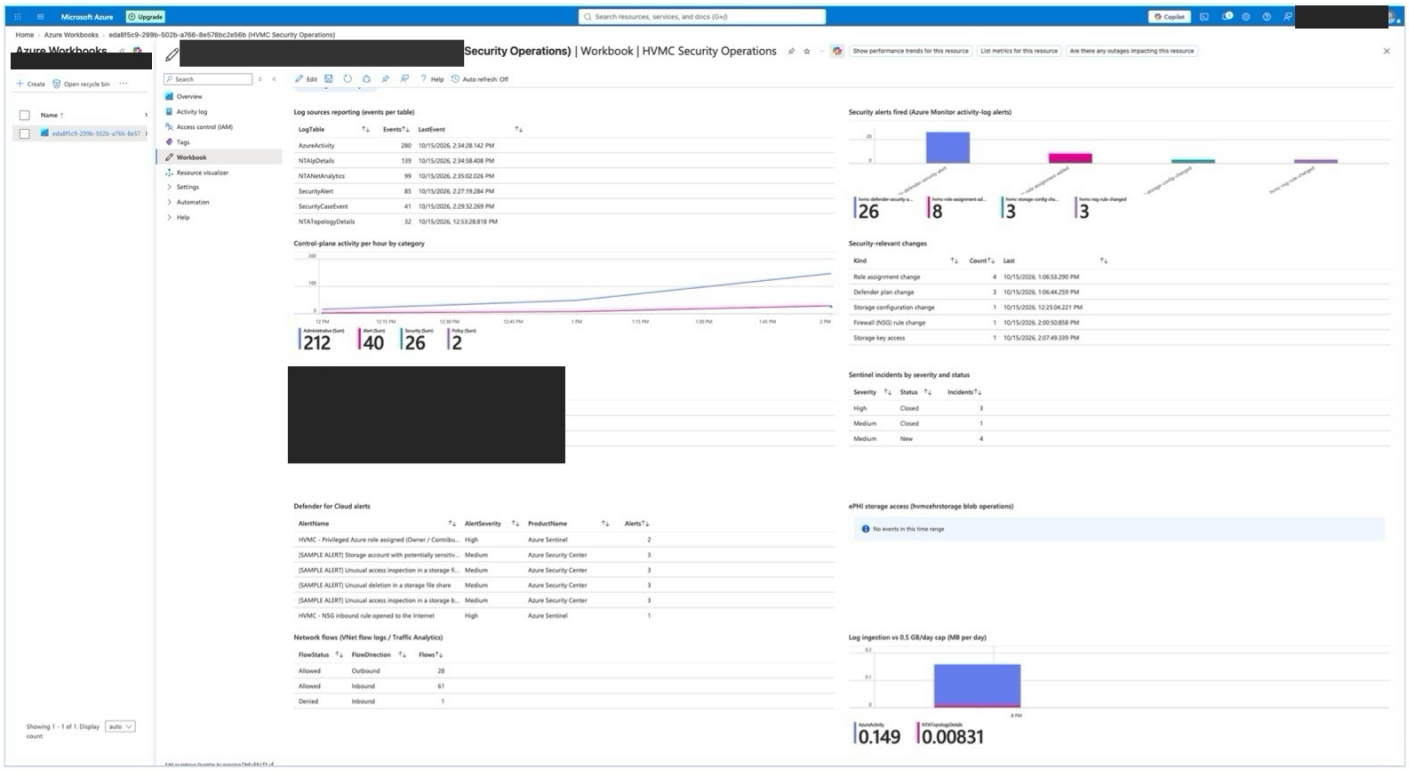



*Figure 21. Azure Workbook 'HVMC Security Operations' in the portal, all ten tiles (Last 7 days): log sources, alerts per rule, activity per hour, security changes, callers, Sentinel incidents, Defender alerts, ePHI access, network flows (61 inbound / 28 outbound allowed, 1 denied) and ingestion vs cap. Captured before the duplicate-incident triage, so 4 Medium incidents still show as New (closed at 18:44 UTC - see section 5).*

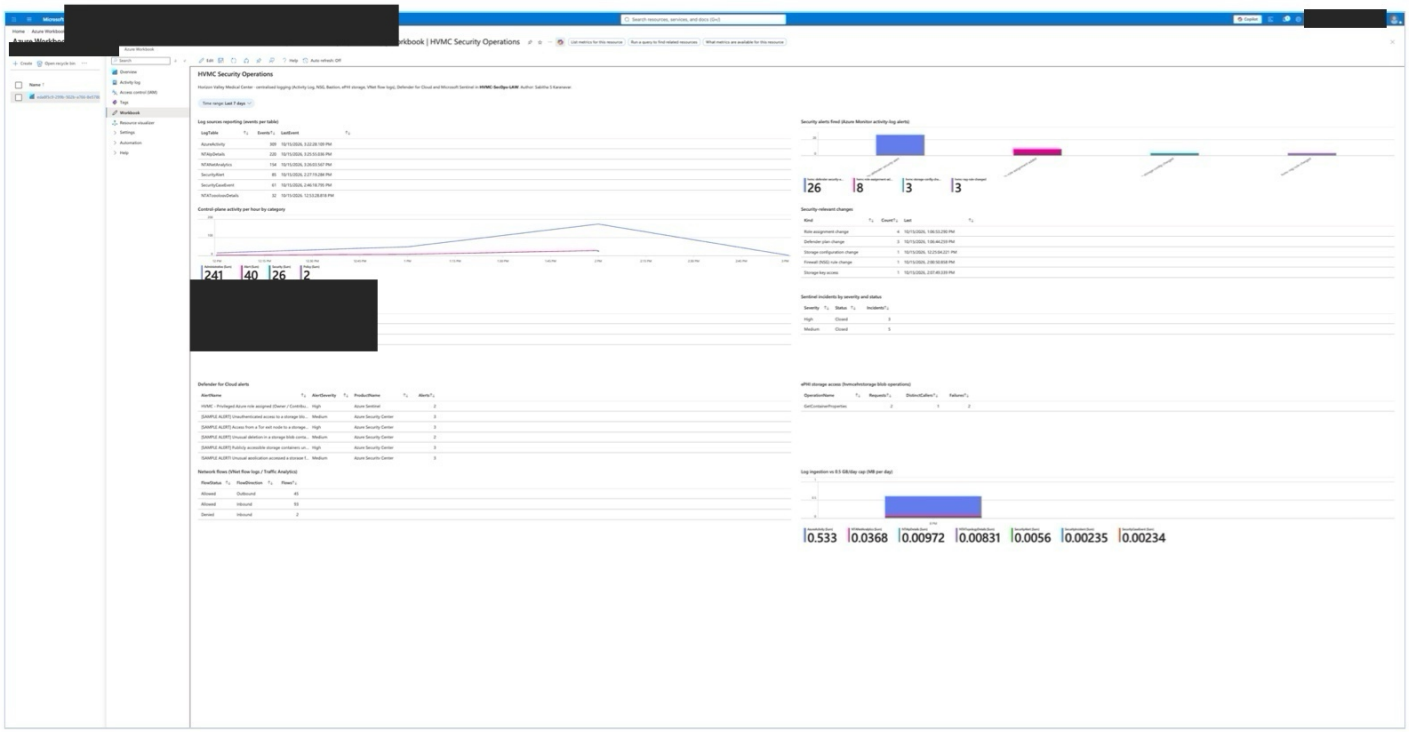



*Figure 22. The same Workbook after triage: Sentinel incidents High 3 Closed and Medium 5 Closed (none New); flows 93 inbound / 45 outbound allowed, 2 denied; first ePHI storage access records.*

**Triage of the first ePHI storage access records (Workbook, 19:11 UTC):** two `GetContainerProperties` requests with status 404 on `hvmcehrstorage`. Investigation in StorageBlobLogs showed authentication type *TrustedAccess*, user agent `SRP/1.0` (the Azure Storage Resource Provider) checking for the Microsoft Purview system container

`$purview`, which does not exist. No blob was listed or read and no identity of HVMC staff was involved, so the activity was classified as benign platform behaviour. This confirms the ePHI audit log (HIPAA §164.312(b)) records even platform-level access.

## 5. Step 4 — Incident response playbook and testing

The full playbook is a separate document, **HVMC_Cloud_IR_Playbook.pdf** (version 1.0). It defines roles and a RACI, a four-level severity scale with response targets, HIPAA breach-notification steps, a common five-phase procedure (identification, containment, investigation, eradication, recovery) and four scenario playbooks, each tied to the detections built in Steps 1–2:

| Playbook | Scenario | Detections that trigger it | Risks | How it was tested |
|---|---|---|---|---|
| **PB-01** | Internet exposure / unauthorised firewall change | HVMC-NSG-Rule-Changed/-Deleted, Public-IP-Created; Sentinel “NSG opened to the Internet” | R03, R10 | Live test |
| **PB-02** | Privilege escalation / privileged role misuse | HVMC-Role-Assignment-Added; Sentinel “Privileged Azure role assigned”; Defender for Resource Manager | R04, R07 | Tabletop exercise |
| **PB-03** | ePHI storage threat (malware, unusual access, key misuse) | Defender for Storage alerts; HVMC-Storage-Keys-Listed; Sentinel “ePHI storage keys retrieved” | R02, R05, R11 | Live test |
| **PB-04** | Security logging disabled or tampered | HVMC-Diagnostic-Setting-Deleted; Sentinel “Security logging disabled” | R01 | Scheduled (Phase 2) |

**Test design principle:** no test weakened a security control. PB-03 used Microsoft Defender for Cloud's built-in sample alerts; PB-01 used a real, approved *hardening* change (an explicit Deny-Internet-inbound rule on the Data-tier NSG, which also treats risk R03); PB-02, whose live version would require granting excess privilege, was run as a tabletop exercise on real alert data.

### 5.1 Live test results (exercise HVMC-IRX-2026-10, 15 Oct 2026, times UTC)

| Test | Triggered | Detected | Time to detect (min) | Detect → contain (min) | Detect → recover (min) | Phases passed |
|---|---|---|---|---|---|---|
| **PB-03** | 18:03:31 | 18:03:32 | 0.0 | 4.3 | 4.7 | 6/6 |
| **PB-01** | 18:00:49 | 18:07:26 | 6.6 | 0.2 | 0.3 | 4/4 |

#### PB-03: ePHI storage threat - Defender for Storage alerts (built-in sample alerts)

| Phase | Done | Result | Evidence |
|---|---|---|---|
| 1 Identification | 18:07:24 | Pass | 26 Defender alerts: [SAMPLE ALERT] Access from a Tor exit node to a storage blob container; [SAMPLE ALERT] Access from a Tor exit node to a storage file share; [SAMPLE ALERT] Access from a suspicious IP to a storage blob container; [SAMPLE ALERT] Access from a suspicious IP to a storage file share; |
| 1b Triage / acknowledge | 18:07:48 | Pass | 26 alerts set to In progress (acknowledged) |
| 2 Containment | 18:07:50 | Pass | secondary access key of hvmcehrstorage regenerated (key2 not used by any application); network posture recorded: {'defaultAction': 'Allow', 'publicNetworkAccess': 'Enabled', 'sharedKey': True} |
| 3 Investigation | 18:07:52 | Pass | alert entities reviewed; ePHI blob data-plane operations in 24 h: 0; control-plane events on the account in 24 h: 0 (all by known callers) |

| | | | |
|---|---|---|---|
| 4 Eradication | 18:07:52 | Pass | Defender for Storage Standard and malware scanning True confirmed; sample alerts carry no payload, so no blob needed removal |
| 5 Recovery | 18:08:15 | Pass | 26 Defender alerts resolved; protection verified still enabled |

### PB-01: Unplanned firewall change on the Data tier - detection and change verification

| Phase | Done | Result | Evidence |
|---|---|---|---|
| 1 Identification | 18:07:39 | Pass | Azure Monitor alert HVMC-NSG-Rule-Changed fired at 2026-10-15T18:07:26Z; e-mail sent to the Security Lead |
| 2-3 Triage, containment decision and investigation | 18:07:41 | Pass | who: [redacted] what: Inbound Deny from Internet to port *; matches approved change HVMC-CHG-04 -> authorised; effect reduces exposure |
| 4 Eradication (exposure sweep) | 18:07:44 | Pass | sweep of all NSGs: no Internet-facing allow rule except the approved Web HTTPS 443 rule |
| 5 Recovery (baseline updated) | 18:07:45 | Pass | baseline updated with the approved change; difference from previous baseline: ['Deny-Internet-Inbound-Data p4000 Deny Internet->*']; App->Data 1433 rule unchanged |

### Microsoft Sentinel incidents raised and closed

| # | Title | Severity | Created | Closed | Minutes open | Classification |
|---|---|---|---|---|---|---|
| #1 | HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator) | High | 17:17:18 | 18:14:21 | 57.0 | BenignPositive / SuspiciousButExpected |
| #2 | HVMC - Privileged Azure role assigned (Owner / Contributor / User Access Administrator) | High | 17:22:19 | 18:14:18 | 52.0 | BenignPositive / SuspiciousButExpected |
| #3 | HVMC - NSG inbound rule opened to the Internet | High | 18:12:22 | 18:14:15 | 1.9 | FalsePositive / IncorrectAlertLogic |
| #4 | HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction | Medium | 18:15:05 | 18:15:39 | 0.6 | BenignPositive / SuspiciousButExpected |
| #5 | HVMC - ePHI storage keys retrieved or storage exposure changed | Medium | 18:17:30 | 18:44:16 | 26.8 | BenignPositive / SuspiciousButExpected |
| #6 | HVMC - NSG rule change on hvmc-data-nsg: rule body not logged - verify access and direction | Medium | 18:20:07 | 18:44:13 | 24.1 | BenignPositive / SuspiciousButExpected |
| #7 | HVMC - ePHI storage keys retrieved or storage exposure changed | Medium | 18:22:36 | 18:44:10 | 21.6 | BenignPositive / SuspiciousButExpected |
| #8 | HVMC - ePHI storage keys retrieved or storage exposure changed | Medium | 18:27:35 | 18:44:06 | 16.5 | BenignPositive / SuspiciousButExpected |

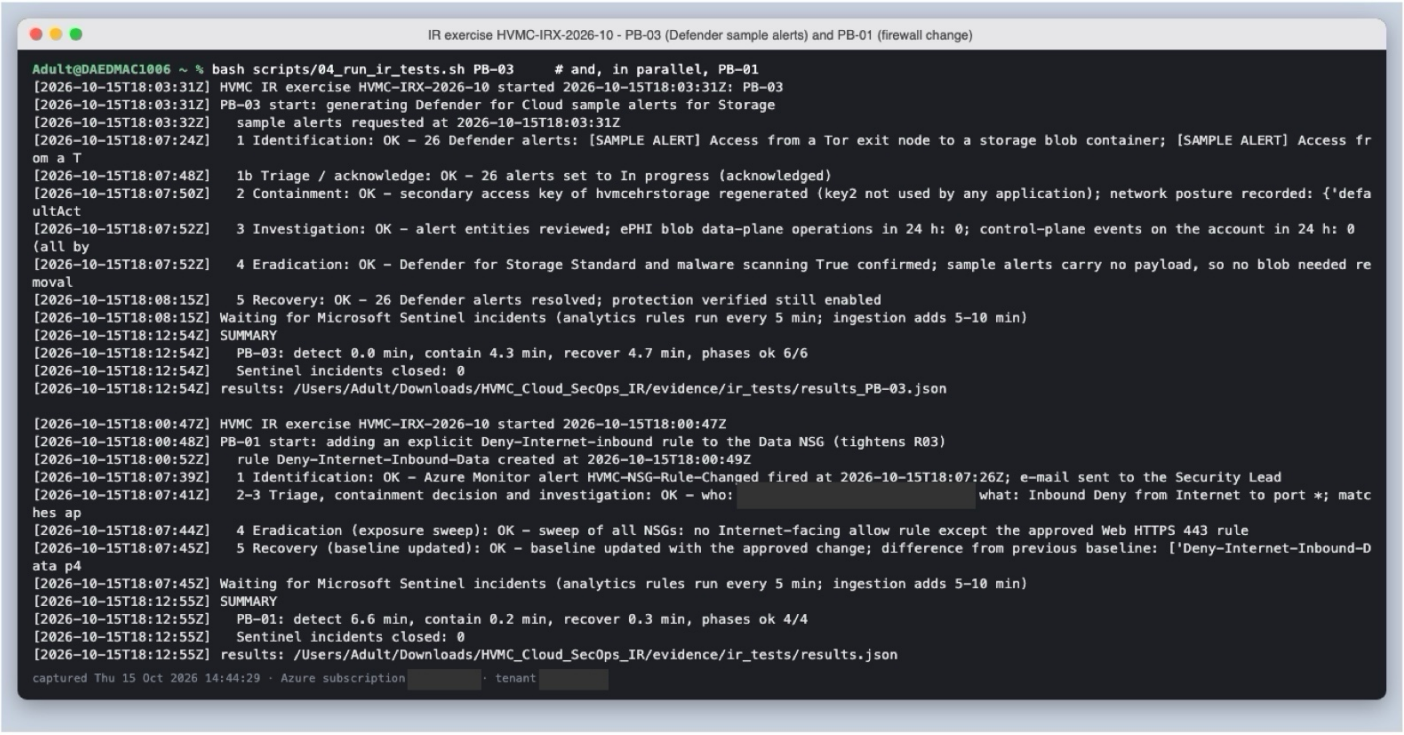



*Figure 23. Live test run log: every playbook phase timestamped.*

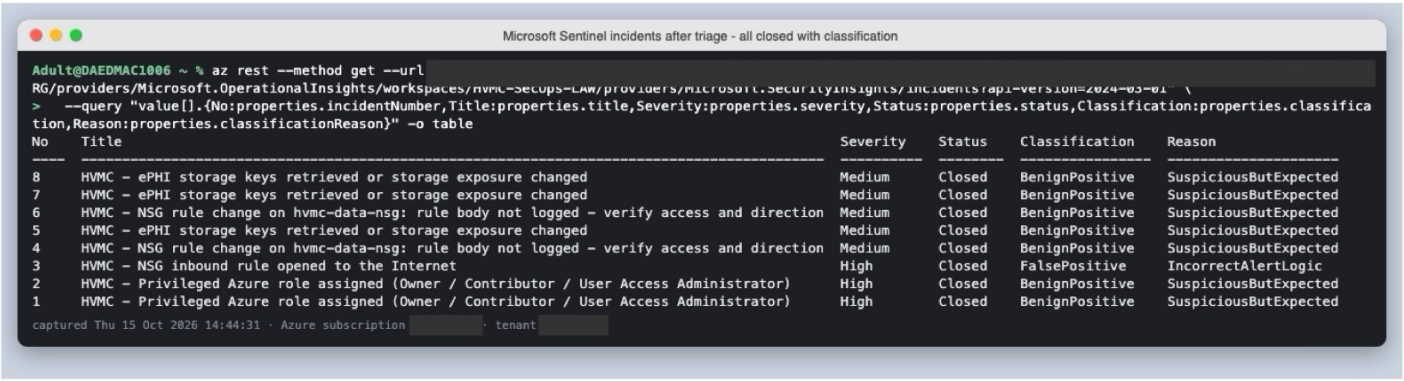



*Figure 24. Sentinel incidents after the exercise: commented and closed with classification.*

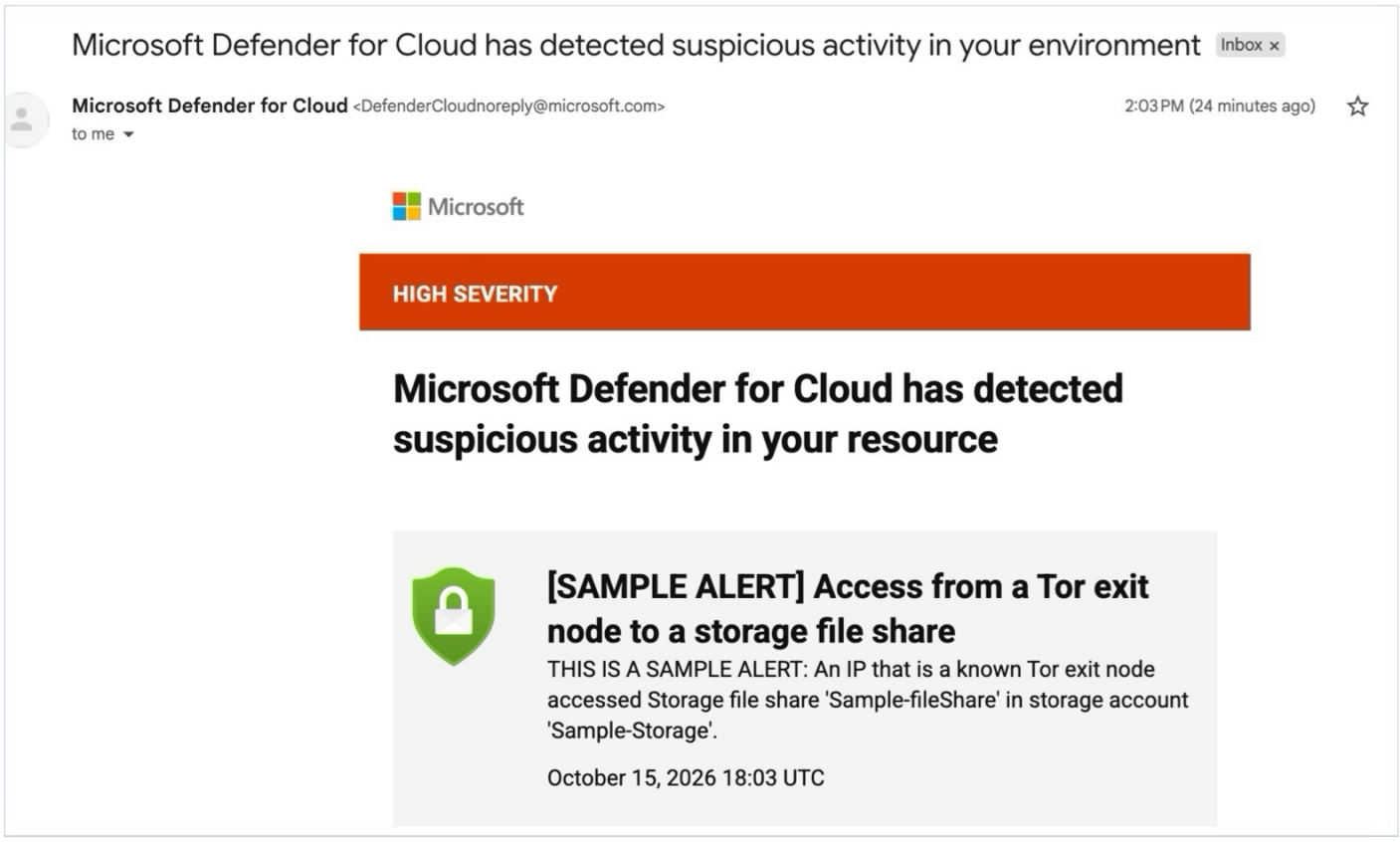



*Figure 25. PB-03 identification: Defender for Cloud e-mail to the Security Lead - High severity sample alert, 18:03 UTC.*

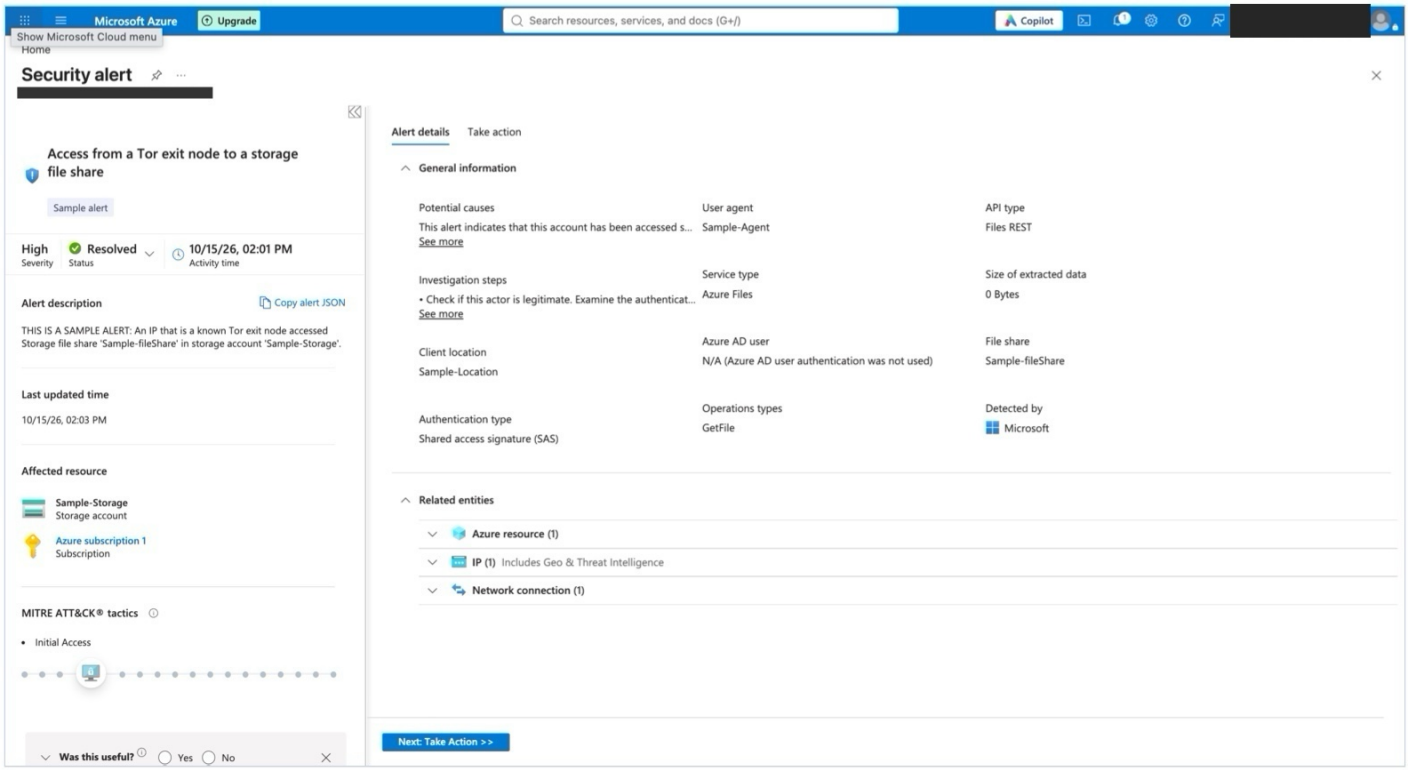



*Figure 26. PB-03 recovery: the same alert in the Azure portal, status Resolved, with MITRE tactic, entities and investigation steps.*

## 5.2 Tabletop exercise — PB-02

Run on real data: the eight HVMC-Role-Assignment-Added alerts of 17:12 UTC were the inject. The team identified them as platform-managed identities in 6 minutes, then rehearsed a variant (User Access Administrator granted to the

clinical test account) through containment, investigation, eradication and recovery. Findings and the improvement plan are in sections 7–8 of the playbook.